HKUST – Department of Computer Science and Engineering
COMP 2711: Discrete Math Tools for Computer Science

Fall 2023 Final Examination

Date: 19 December 2023 Time: 12:30–15:30

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This is a closed book exam. It consists of **11** questions. You are allowed to bring an A4 one-page double-sided handwritten cheat sheet.
- Please write your name, student ID, email, lecture section in the space provided at the top of this page.
- Please sign the honor code statement on page 2.
- Answer all the questions within the space provided on the examination paper. You may also use the back of the pages. Each question is on a separate page. This is for clarity and is not meant to imply that each question requires a full-page answer. Some can be answered using only a few lines.
- Solutions can be written in terms of binomial coefficients, factorials, and the $C(n,k)$, $P(n,k)$ notation. For example, you can write $\binom{5}{3}+\binom{4}{2}$ instead of 16. Avoid using nonstandard notations such as ${}_nP_k$ or ${}_nC_k$. Calculators are not allowed for the exam.

Problems	1	2	3	4	5	6	7	8	9	10	11	Total
Points	5	10	7	13	11	9	11	11	12	6	5	100
Score												

As part of HKUST's introduction of an honor code, the HKUST Senate has recommended that all students be asked to sign a brief declaration printed on examination answer books that their answers are their own work, and that they are aware of the regulations relating to academic integrity. Following this, please read and sign the declaration below.

I declare that the answers submitted for this examination are my own work.

I understand that sanctions will be imposed, if I am found to have violated the University regulations governing academic integrity.

Student's Name: _____

Student's Signature: _____

Problem 1: Logic [5 points]

In this question the domain is the set of all integers. Consider the following predicates:

$P(x)$: x is prime.

$Q(x, y)$: x, y are relatively prime (co-prime).

$E(x, y, z)$: x, y are congruent mod z (i.e., $x \equiv y \pmod{z}$).

(a) Determine whether the following logical expressions are True or False. **No justification is needed**. ($a \mid b$ means b is divisible by a)

(i) **[1 point]** $\forall x \forall a \forall b : x \mid ab \rightarrow (x \mid a \ \vee \ x \mid b)$

(ii) **[1 point]** $\forall x \forall a \forall b : P(x) \rightarrow (x \mid ab \rightarrow (x \mid a \ \vee \ x \mid b))$

(iii) **[1 point]** $\forall x \forall a \forall b : (x \mid ab \rightarrow (x \mid a \ \vee \ x \mid b)) \rightarrow P(x)$

(b) **[2 points]** Write down Fermat's little theorem using quantifiers and the predicates given above. You are also allowed to use power. As an example $E(x^2, y, z)$ is acceptable.

(**Fermat's little theorem:** If p is prime and a is any integer not divisible by p, then $a^{p-1} \equiv 1 \pmod{p}$.)

Answer: (a) (i) False

(ii) True

(iii) False

(b) $\forall x \forall y : (P(x) \wedge Q(y, x)) \implies E(y^{x-1}, 1, x)$.

Marking scheme: (a) (i) +1 point for correct answer

(ii) +1 point for correct answer

(iii) +1 point for correct answer

(b) +2 points for Precise answer

- Deduction: -1 point for missing quantifier

Problem 2: Counting and Probability [10 points]

(a) There are 5 blue balls and 3 green balls in a basket. Every time, we close our eyes and pick a ball at random from the basket (and we will not put the ball back in the basket after picking it).

(i) **[3 points]** What is the probability that we see a green ball for the first time in our 4th pick? Show the steps to calculate the final answer as a single fraction like $\frac{a}{b}$.

(ii) **[4 points]** What is the probability that we see a green ball in our 4th pick? You can leave the final answer as summation of several fractions that might include combinations as well.

(b) **[3 points]** How many 4-digit positive integers a exists such that both a and $2 \times a$ have only even digits? For example, 2004 is a valid number, but $a = 4282$ is not because $2a = 2 \cdot 4282 = 8564$ which contains the odd digit "5". **(Notice that the <u>leftmost</u> digit in a 4-digit number <u>cannot</u> be 0)**

Answer: (a) (i) In order to see a green ball for the first time in 4th pick, then the first 3 balls must be blue. So it is equal to $\frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} \cdot \frac{3}{5} = \frac{3}{28}$

(ii) There are 3 cases. We calculate each separately and add them together at the end.

- **Case 1:** If that green ball is the first green ball we see. This is the same as part (i) which is $\frac{3}{28}$.
- **Case 2:** If that green ball is the second green ball we see. This means in the first 3 balls we have to see exactly 2 blue balls and 1 green ball. the probability here is: $\frac{\binom{5}{2} \cdot \binom{3}{1}}{\binom{8}{3}}$. So in this case the probability would be $\frac{\binom{5}{2} \cdot \binom{3}{1}}{\binom{8}{3}} \times \frac{2}{5}$
- **Case 3:** If that green ball is the third green ball we see. This means in the first 3 balls we have to see exactly 1 blue balls and 2 green ball. the probability here is: $\frac{\binom{5}{1} \cdot \binom{3}{2}}{\binom{8}{3}}$.

  So the probability would be $\frac{\binom{5}{1} \cdot \binom{3}{2}}{\binom{8}{3}} \times \frac{1}{5}$

So the final answer is $\frac{3}{28} + \frac{\binom{5}{2} \cdot \binom{3}{1}}{\binom{8}{3}} \cdot \frac{2}{5} + \frac{\binom{5}{1} \cdot \binom{3}{2}}{\binom{8}{3}} \cdot \frac{1}{5}$

(b) Notice that when we have a carryover thing it changes the oddity of the next digit. So for example if we have a valid number a like $a = 2822$. When we multiply it by 2 we get the invalid number 5644 and that is because we got a carryover when multiplying 8 by 2. So in order to have a valid number a such that $2a$ is also valid, they should not

produce any carryover value. Which means they can be either $0, 2, 4$. So the number $\overline{abcd}$ can have $2 \times 3 \times 3 \times 3 = 54$ possible options.

Marking scheme: (a) (i) +3 points for correct answer

(ii) +1 point for each case;
+1 point for the final answer;
Deduction: -1 point for wrong case. -1 point for wrong answer. -1 point for wrong computation.

(b) +3 points for the totally correct answer.
Deduction: -1 point for the answer with wrong digit.

Problem 3: Conditional Probability [7 points]

Suppose that one person in 10,000 people has a rare genetic disease. There is an excellent test for the disease; 99.9% of people with the disease test positive and only 0.02% who do not have the disease test positive. What is the probability that someone who tests negative does not have the disease? You have to show all the steps. But you do not have to calculate the final numerical values in each step.

Answer: First, we determine these events: A is the event that someone has the disease, and B is the event that someone's test is positive. From the problem we know:

$P(A) = \frac{1}{10000}$
$P(B|A) = 0.999$
$P(B| \sim A) = 0.0002$

$P(B) = P(A) * P(B|A) + P(\sim A) * P(B| \sim A) = \frac{1}{10000} \cdot 0.999 + (1 - \frac{1}{10000}) * 0.0002 =$

What is the probability that someone who tests negative does not have the disease? So we want to calculate : $P(\sim A| \sim B) = \frac{P(\sim B|\sim A)P(\sim A)}{P(\sim B)}$ in which $P(\sim B| \sim A) = 1 - P(B| \sim A)$ and $P(\sim B) = 1 - P(B)$ and $P(\sim A) = 1 - P(A)$.

Marking scheme: -1 point for each wrong partial probabilities
-4 points for minor logical error

Problem 4: Random Variables [13 points]

The CSE Department of HKUST has a total of n light bulbs and m switches. We know that each light bulb is connected to at least one switch. By flipping a switch, the status of every light bulb connected to it will change (so if it was OFF it will be turned ON and vice versa). Assume X_i is the random variable that shows the status of the ith light bulb. $X_i = 0$ means light bulb i is OFF and $X_i = 1$ means it is ON. Also, let Y be the random variable that shows the number of light bulbs that are ON in the department. Note that the initial status of every light bulb is arbitrary.

(a) **[3 points]** Imagine you flip every switch with 50% probability. What is the expected value of each $X_1, X_2, X_3, \cdots, X_n$? **Briefly** explain why.

(b) **[3 points]** What is the expected value of Y? **Briefly** explain why.

(c) **[3 points]** Assume K is a random variable. Why is it not possible that $K > E(K)$ for all possible values of K? In other words, explain why there must be values for K such that $K \leq E(K)$.

(d) **[4 points]** Recall the CSE Department with its light bulbs and switches again. Prove that it is possible to have at least half of the light bulbs being turned ON.

(**Hint:** You might want to use some facts from the previous parts. You are also allowed to use the fact in part (c) even if you could not prove it)

Answer:

(a) The expected value of $E(X_i) = 0.5$ for any $i \in \{1, 2, \cdots, n\}$. Because each light bulb is on in half of all possible ways of switches.

(b) $E(Y) = E(X_i) + E(X_2) + \cdots + E(X_n) = \frac{n}{2}$.

(c) It is not possible because if all possible values for K are bigger than $E(K)$ then the expected value should be larger than its current value.

(d) If we flip every switch with 50% probability, then on expectation half of the light bulbs will be on. So based on (c) there exists a way for switches such that at least half of the light bulbs are on.

Marking scheme:

(a) 3 points

(b) 1 points for $E(Y) = E(X_i) + E(X_2) + \cdots + E(X_n)$, 2 points for the final answer

(c) 1 point only if the proof is not general (i.e. use an example to prove)

(d) 4 points in total, 2 points for implying (c)

Problem 5: RSA [11 points]

This problem is on the RSA algorithm for public key cryptogtaphy. To generate his keys, Bob samples two prime numbers $p = 17, q = 19$. So, $n = pq = 323$ and $T = 288$.

(a) **[5 points]** Assume that Bob chooses $e = 23$. Compute his private key d by running the extended GCD algorithm.

(b) **[6 points]** Suppose Alice wants to encrypt message $m = 18$ using Bob's public key (n, e). Use the repeated squaring technique to calculate the ciphertext $c = m^e$.

Answer: (a) Find the inverse of $e = 23$ in Z_{288}:

$$\begin{aligned} 288 &= 23 \cdot 12 + 12 \\ 23 &= 12 \cdot 1 + 11 \\ 12 &= 11 \cdot 1 + 1 \\ 11 &= 1 \cdot 11 + 0 \end{aligned}$$

Hence we get:

$$\begin{aligned} 1 &= 12 - 11 \cdot 1 \\ &= 12 - (23 - 1 \cdot 12) \cdot 1 \\ &= 12 \cdot 2 - 23 \cdot 1 \\ &= (288 - 23 \cdot 12) \cdot 2 - 23 \cdot 1 \\ &= 288 \cdot 2 + 23 \cdot (-25) \end{aligned}$$

The private key $d = -25 \equiv 263 \pmod{288}$

(b)

$$\begin{aligned} 18^{2^0} \bmod 323 &= 18 \\ 18^{2^1} \bmod 323 &= 18^2 \bmod 323 = 1 \\ 18^{2^2} \bmod 323 &= 1^2 \bmod 323 = 1 \\ 18^{2^3} \bmod 323 &= 1^2 \bmod 323 = 1 \\ 18^{2^4} \bmod 323 &= 1^2 \bmod 323 = 1 \end{aligned}$$

$23 = 2^4 + 2^2 + 2^1 + 2^0$

$$\begin{aligned} 18^{23} &\equiv 18^{2^4} \cdot 18^{2^2} \cdot 18^{2^1} \cdot 18^{2^0} \\ &\equiv 1 \cdot 1 \cdot 1 \cdot 18 \\ &\equiv 18 \pmod{323} \end{aligned}$$

Marking scheme: (a) +5 points for correct answer.
Deduction: -2 points if mis-calculation at the end (25 instead of -25)

(b) +2 points if the remainder of each component is calculated.
+4 points for the rest of 3 points if the final result correctly calculated by multiply these products.

Problem 6 Number Theory, Modular Arithmetic [9 points]

(a) **[6 points]** Let a, b, c, d be integers. Show that the product

$$(a-b)(a-c)(a-d)(b-c)(b-d)(c-d)$$

is divisible by 12.

(**Hint:** You might want to prove that there exist at least two numbers among a, b, c, d such that they have the same remainder mod 3 and this might help you with the main proof)

(b) **[3 points]** What is the remainder of 17^{2023} mod 14?

Answer: (a) Out of 4 different numbers, two of them have the same remainder mod 3 based on pigeonhole principle. So one of those six elements are divisible by 3. Now out of these four numbers, either we have two odd and two even or there exists at least three numbers with same oddity. In each case we can get two different pair of numbers that their difference is an even number. So it would be dividable by 4. So the total would be dividable by 12.

(b) $17 \equiv 3 \pmod{14}$ So we have: $3^3 \equiv 27 \equiv -1 \pmod{14}$
$(3^3)^{674} \equiv 3^{2022} \equiv (-1)^{674} \equiv 1 \pmod{14}$ $3^{2023} \equiv 3 \cdot 3^{2022} \equiv 3 \cdot 1 \equiv 3$.

Marking scheme: (a) +3 points for the product is divisible by 3;
+3 points for the product is divisible by 4.

(b) +3 points for correct answer with an acceptable explanation.
+1 point for correct answer without explanation.

Problem 7: Chinese Remainder [11 points]

Solve this system of linear congruences:

$$\begin{aligned} 3x &\equiv 1 \pmod{10} \\ 5x &\equiv 3 \pmod{6} \end{aligned}$$

Show all the steps.

(**Hint:** You might want to get rid of the coefficients somehow first)

Answer: One approach is to simplify each equation first by multiplying each side by the inverse of the used coefficient.

In first equation the inverse of 3 mod 10 exists as they are relatively prime and the inverse is 7.

In second equation the inverse of 5 mod 6 exists as they are relatively prime and the inverse is 5. So the equation converts to :

$$\begin{aligned} x &\equiv 7 \pmod{10} \\ x &\equiv 3 \pmod{6} \end{aligned}$$

Now we can not use CRT directly as $6, 10$ are not relatively prime. So, we use the same idea as in homework to break them down them into separate cases that are pairwise relatively prime. Which means we need to solve:

$$\begin{aligned} x &\equiv 2 \pmod{5} \\ x &\equiv 1 \pmod{2} \\ x &\equiv 0 \pmod{3} \end{aligned}$$

$m = 30 \quad M_1 = 6 \quad M_2 = 15 \quad M_3 = 10.$

$y_1 = 1, y_2 = 1, y_3 = 1$

$x = 2 \cdot 6 \cdot 1 + 1 \cdot 15 \cdot 1 + 0 \cdot 10 \cdot 1 = 27.$

$x \equiv 27 \pmod{30}$

Marking scheme: +2 points for $x \equiv 7 \bmod 10$
+2 points for $x \equiv 3 \bmod 6$
+2 points for $x \equiv 1 \bmod 2$, $x \equiv 2 \bmod 5$, $x \equiv 0 \bmod 3$ +5 points for calculating by CRT and get the correct result $x \equiv 27 \bmod 30$

Problem 8: Recurrence and Induction [11 points]

(a) **[7 points]** For every non-negative integer n, a_{n+1} is calculated from a_n with the following rules:
If the **rightmost** digit of a_n is bigger than 5, then $a_{n+1} = 9 \times a_n$. Otherwise, we just "remove" the rightmost digit of a_n and the remaining number is a_{n+1}. If a_{n+1} does not contain any digits we stop.

As an example, if $a_n = 124$ then $a_{n+1} = 12$ as a_n's rightmost digit is 4 and we simply remove it from 124 to form 12. Also if $a_n = 16$ then $a_{n+1} = 9 \times 16 = 144$ as a_n's rightmost digit was bigger than 5.

Use strong induction on a_0 to prove that this process stops in **finitely** many steps for any arbitrary a_0. Please write a complete induction proof consisting of base case or cases, inductive hypothesis, inductive step and conclusion.
(**Hint 1:** If this process stops for any $1 \leq a_0 \leq k$, can you show it will stop in finitely many steps for $a_0 = k + 1$ as well?)
(**Hint 2:** If the rightmost digit of a number a_n is larger than 5, what can we say about the possible values for the rightmost digit of $9 \times a_n$?)

(b) **[4 points]** Calculate the closed form of the following recursive relation:

$$a_0 = 1, \quad a_n = 3a_{n-1} + 2 \quad (\text{for } n \geq 1)$$

Answer:

(a) We use strong induction on the first element a_0. Obviously this process stops in 1 step if $a_0 = 1$. Assume that this assumption is correct for all values between 1 up to k. And now we prove it will be also correct for $k+1$. If $K+1$ ends with a digit that is smaller or equal than 5 then we just simply remove that digit and the result number is already smaller than $K+1$ so it will stop after finitely many steps. If $K+1$ ends with a digit that is bigger than 5, then We get $9(k+1)$ as the next number that definitely ends with a digit smaller than 6. So for its next step we need to remove the rightmost digit that will give us a number that is again smaller than $k+1$. (because $9(k+1)/10$ is smaller than $k+1$ for sure)

(b) $a_0 = 1$ $b = 3$ $c = 2$
$f_n = 1 \cdot 3^n + 2\frac{3^n - 1}{3-1}$

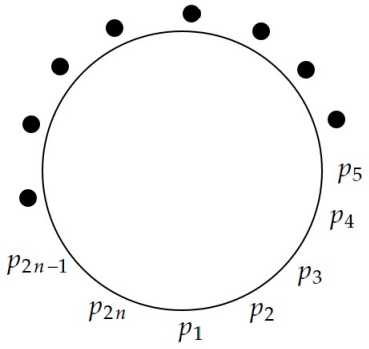
Marking scheme: (a) +2 points for Base Case;
+1 point for Induction Hypothesis;
+2 points for case with last digit less and equal to 5;

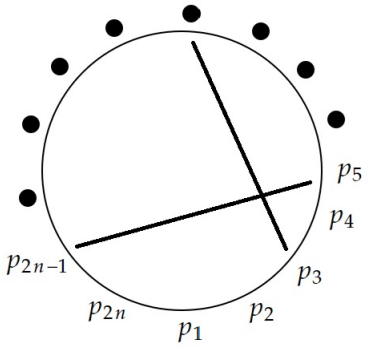
+2 points for case with last digit greater than 5 provided that the a_{n+2} is less than k to get full points of 2.

(b) +2 points for deriving the first part of a_n;
+2 points for deriving the geometric progression provided that you must simplified the geometric progression.

Problem 9: Recursive [12 points]

Assume that there are $2n$ people sitting around a round table. We want to split them into n pairs to engage in handshakes so that no arms cross, given that **nobody** is allowed to shake hands with more than one person at once. Let a_{2n} denote the number of ways that we can pair them legally without any arm crossing. You can see an invalid handshake in the figure below. Please answer the following questions by referring to the figure:





Invalid, this is an arm cross

(a) **[2 points]** Is it possible to pair p_1, p_5 together for a handshake? Please explain why in at most 3 lines.

(b) **[2 points]** Assume that you want to first pair p_1 with someone. List all potential candidates who can be paired with p_1 initially?

(c) **[5 points]** Assume p_1 is paired with p_j initially. Derive the number of ways to pair up the remaining people in terms of a_i.

(**Hint:** Notice that this handshake will create two separate groups of people on the "left" side and the "right" side of it)

(d) **[3 points]** Show the recursive relation of a_{2n}. **You do not have to represent it in closed form.** You can assume $a_0 = 1$.

Answer: (a) No, because then on their right side they have $\{p_2, p_3, p_4\}$ which one of them has be paired with someone from the other side and this cause cross arm.

(b) It can only be paired with p_j where j is even. Otherwise the number of people in one side is odd and will guarantee an arm cross.

(c) In this case we have one group of size $j-2$ on one side and a group of size $2n-j$ on the other side. So We have $a_{j-2} \cdot a_{2n-j}$ number of ways to pair the remaining people in a valid way.

(d) We just need to calculate the summation in which the first person is paired with $p_2, p_4, p_6, \cdots, p_{2n}$.

Marking scheme: (a) +1 point for correct answer;
+1 point for explanation.

(b) +2 points for showcasing all potential even numbers;
Deduction: -1 point for each missing or partially presented even number.

(c) +1 point for illustrating it is divided into groups of j-2 and +1 point for 2n-j;
+2 points for showing correct steps in deriving your solution;
+1 point for correct answer.

(d) +2 point for showing correct steps in deriving your solution;
+1 points for correct answer.

Problem 10: Algorithm and Complexity Analysis [6 points]

Analyze the worst-case time complexity of the following algorithm for finding the first term of a sequence of integers equal to some previous term.

procedure find duplicate($a_1, a_2, \ldots, a_n$: integers)
location := 0 {no match found yet}
$i := 2$
while $i \leq n$ and *location* $= 0$
  $j := 1$
  while $j < i$ and *location* $= 0$
    if $a_i = a_j$ **then** *location* := i
    else $j := j + 1$
  $i := i + 1$
return *location*
{location is the subscript of the first value that repeats a previous value in the sequence and is 0 if there is no such value}

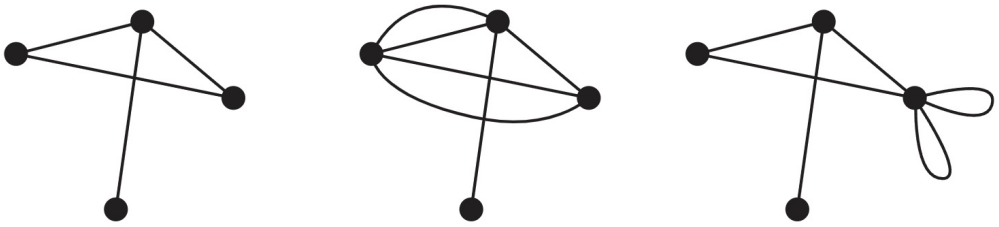
Answer: The worst case is that in which we do not find any term equal to some previous term. In that case, we need to go through all the terms a_2 through a_n , and for each of those, we need to go through all the terms occurring prior to that term. Thus the inner loop of our algorithm is executed once for $i = 2$ (namely for $j = 1$), twice for $i = 3$ (namely for $j = 1$ and $j = 2$), three times for $i = 4$, and so on, up to $n - 1$ times for $i = n$. Thus the number of comparisons that need to be made in the inner loop is $1 + 2 + 3 + \cdots + (n - 1)$. This sum is $(n - 1)(n - 1 + 1)/2$, which is clearly $O(n^2)$ but no better. Bookkeeping details do not increase this estimate.

Marking scheme:
+6 points for precise conclusion of $O(n^2)$ with Correct steps/formula
+4 points for precise conclusion of $O(n^2)$ with incorrect or missing steps/formula
+4 points for correct formula of n(n-1)/2 without precise conclusion
+0 point Otherwise

Problem 11: Graph [5 points]

Give a contrapositive proof that if there are **exactly** two odd degree vertices in a simple graph, then there is a path between them. In other words, those two nodes have to be in the same connected component.

(Recall that a simple graph is an undirected, unweighted graph such that there are no multiple edges between any two nodes and contains no loops. It can be connected or disconnected though. The figure below shows some examples for simple and nonsimple graphs.)



simple graph | *nonsimple graph with multiple edges* | *nonsimple graph with loops*

Answer: Consider a graph with exactly two nodes v, u of odd degree. We prove by contrapositive. So v, u must be in two disconnected part of graph G called G_v, G_u respectively. Otherwise there was a path joining them. Consider G_v or G_u. This subgraph is still a graph and the number of nodes with odd degree has to be even in each because the sum of vertices degrees are always even number. So there exists another vertex v' in G_v with odd degree. This contradicts our initial assumption that u, v are the only vertices with odd degree.

Marking scheme: +5 points for any logical proof that illustrates the solution utilized contrapositive statements;
Deduction: -1 point for any cases without giving contrapositive proof.

Draft Paper

Draft Paper

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